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Inventor(s)	OBUKURO; YUUJI et al.

CARTRIDGE, IMAGE FORMING APPARATUS, DEVELOPING APPARATUS, AND TONER CARTRIDGE

Abstract

A cartridge is provided that includes a first accommodating portion configured to accommodate toner, a first rotation shaft configured to be rotatable about a first rotation axis line, a first flexible member fixed to the first rotation shaft, a first magnetized member provided on the first flexible member, a second accommodating portion configured to accommodate toner to be supplied to the first accommodating portion and being aligned with the first accommodating portion in a first direction, a second rotation shaft configured to be rotatable about a second rotation axis line, a second flexible member fixed to the second rotation shaft, a second magnetized member provided on the second flexible member, and a magnetism sensing sensor provided between the first rotation axis line and the second rotation axis line in the first direction, and configured to sense magnetism in a sensing-capable range.

Inventors: **OBUKURO; YUUJI (Saitama, JP), HIJIKATA; SHUNSUKE (Kanagawa, JP)**

Applicant: **CANON KABUSHIKI KAISHA** (Tokyo, JP)

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electrophotographic image forming apparatus such as a photocopier, printer, or the like, which employ an electrophotographic system, a developing apparatus which is used in the electrophotographic image forming apparatus, and a cartridge which is capable of being attached to and detached from the electrophotographic image forming apparatus.

Description of the Related Art

[0002] In an image forming apparatus that employs the electrophotographic system, a surface of an electrophotographic photosensitive member (hereinafter referred to as “photosensitive drum”) is uniformly charged by charging means, and an electrostatic latent image is formed, with the surface of the photosensitive drum, which is charged, being exposed by exposing means. This electrostatic latent image is then developed by a developing apparatus and a developer image (hereinafter also referred to as “toner image”) is formed of developer (hereinafter also referred to as “toner”) on the photosensitive drum, and this toner image is transferred onto a recording material by transfer means. Thereafter, the toner image is heated by fixing means, and is thus fixed upon the recording material.

[0003] As such an image forming apparatus, an image forming apparatus that is configured such that an amount of developer within a container that holds the developer can be sensed is known. Japanese Patent No. 3971330 and Japanese Examined Utility Model Application Publication No. H 01-032049 disclose a configuration that is equipped with a magnetized member that is magnetized and a sensor for sensing the magnetism thereof, and that senses the amount of developer on the basis of sensing results of the sensor. FIGS. **14** and **15** of the present disclosure illustrate a configuration example of sensing the amount of developer using the magnetized member. FIG. **14** illustrates a toner cartridge **70** that is detachably attachable to an image forming apparatus, this toner cartridge **70** being equipped with a magnetized member **71** that is disposed inside of a toner accommodating portion, and a detecting portion **72** that detects this member. Also, FIG. **15** illustrates a remaining toner amount sensing apparatus **80** (toner hopper) that is equipped with a magnetized member **81** disposed inside of a toner accommodating portion, and a detecting portion **82** that detects this member.

SUMMARY OF THE INVENTION

[0004] An object of the present invention is to provide technology that enables sensing of an amount of developer while conserving space.

[0005] In order to achieve the above object, a cartridge according to the present application is a cartridge that detachably attachable to an image forming apparatus, the cartridge including: [0006] a first accommodating portion configured to accommodate toner; [0007] a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line; [0008] a first flexible member extending in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; [0009] a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end; [0010] a second accommodating portion configured to accommodate toner to be supplied to

the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction; [0011] a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line; [0012] a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; [0013] a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end of the second flexible member, including the second free end; and [0014] a magnetism sensing sensor provided between the first rotation axis line and the second rotation axis line in the first direction, and configured to sense magnetism in a sensing-capable range.
[0015] Also, in order to achieve the above object, an image forming apparatus according to the present application includes: [0016] an apparatus main body; and [0017] a cartridge configured to be detachably attachable to the apparatus main body, the cartridge including [0018] a first accommodating portion configured to accommodate toner, [0019] a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line; [0020] a first flexible member extending in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; [0021] a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end; [0022] a second accommodating portion configured to accommodate toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction; [0023] a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line; [0024] a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; [0025] a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end of the second flexible member, including the second free end; and [0026] a magnetism sensing sensor provided between the first rotation axis line and the second rotation axis line in the first direction, and configured to sense magnetism in a sensing-capable range.
[0027] Also, in order to achieve the above object, an image forming apparatus according to the present application includes: [0028] an image bearing member on which an electrostatic latent image is formed; [0029] a toner bearing member configured to bear toner for developing the electrostatic latent image; [0030] a first accommodating portion that accommodates the toner to be supplied to the toner bearing member; [0031] a first rotation shaft that is provided inside of the first accommodating portion and that is configured to be rotatable about a first rotation axis line; [0032] a first flexible member that extends in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; [0033] a first magnetized member that is provided in a region of the first flexible member between the first fixed end of the first flexible member and the first free end thereof, including the first free end; [0034] a second accommodating portion that accommodates toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction; [0035] a second rotation shaft that is provided inside of the second accommodating portion and that is configured to be rotatable about a second rotation axis line; [0036] a second flexible member that extends in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second

free end opposite to the second fixed end in the direction intersecting the second rotation axis line; [0037] a second magnetized member that is provided in a region of the second flexible member between the second fixed end of the second flexible member and the second free end thereof, including the second free end; and [0038] a magnetism sensing sensor that is provided between the first rotation axis line and the second rotation axis line in the first direction, and that senses magnetism in a sensing-capable range.

[0039] Also, in order to achieve the above object, a developing apparatus according to the present application includes: [0040] a developing member; [0041] a first accommodating portion configured to accommodate toner; [0042] a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line; [0043] a first flexible member extending in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; [0044] a first magnetized member provided in a region between the first fixed end of the first flexible member and the first free end of the first flexible member, including the first free end; and [0045] a magnetism sensing sensor configured to sense magnetism in a sensing-capable range, [0046] wherein the developing apparatus is configured such that a cartridge is detachably attachable, the cartridge including a second accommodating portion accommodating toner to be supplied to the first accommodating portion, a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line, a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line, and a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end of the second flexible member, including the second free end, [0047] wherein the first accommodating portion is aligned with the second accommodating portion in a first direction, in an orientation in which the cartridge is attached to the developing apparatus, and [0048] wherein the cartridge is attached such that the magnetism sensing sensor is located between the first rotation axis line and the second rotation axis line in the first direction.

[0049] Also, in order to achieve the above object, a toner cartridge according to the present application is a toner cartridge configured to be detachably attachable to a developing apparatus that includes a developing member, a first accommodating portion configured to accommodate toner, a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line, a first flexible member extending in a direction intersecting the first rotation axis line and having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line, a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end and a magnetism sensing sensor configured to sense magnetism in a sensing-capable range, the toner cartridge comprising: [0050] a second accommodating portion configured to accommodate toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction in an orientation in which the toner cartridge is attached to the developing apparatus; [0051] a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line; [0052] a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; and [0053] a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end thereof of the second flexible member, including the second free end, [0054] wherein the magnetism sensing sensor is attached to

the developing apparatus so as to be located between the first rotation axis line and the second rotation axis line in the first direction.
[0055] According to the present invention, an amount of developer can be sensed while conserving space.
[0056] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0057] FIG. **1** is a cross-sectional view of a cartridge according to a first embodiment;
[0058] FIG. **2** is a schematic cross-sectional view of an image forming apparatus according to the first embodiment;
[0059] FIG. **3** is a schematic cross-sectional view of the image forming apparatus according to the first embodiment;
[0060] FIG. **4** is a schematic cross-sectional view of the image forming apparatus according to the first embodiment;
[0061] FIG. **5** is a schematic cross-sectional view of the image forming apparatus according to the first embodiment;
[0062] FIG. **6** is a perspective view of the cartridge according to the first embodiment;
[0063] FIG. **7** is a schematic cross-sectional view of a developing container and a toner container according to the first embodiment;
[0064] FIGS. **8**A to **8**C are explanatory diagrams regarding phase matching according to the first embodiment;
[0065] FIGS. **9**A to **9**C are explanatory diagrams regarding a remaining toner amount sensing method according to the first embodiment;
[0066] FIGS. **10**A to **10**C are explanatory diagrams regarding the remaining toner amount sensing method according to the first embodiment;
[0067] FIG. **11** is an explanatory diagram of a relation between amount of time of sensing and remaining toner amount, according to the first embodiment;
[0068] FIG. **12** is an explanatory diagram of a relation between amount of time of sensing and remaining toner amount, according to the first embodiment;
[0069] FIGS. **13**A and **13**B are cross-sectional views of the cartridge according to the first embodiment;
[0070] FIG. **14** is an explanatory diagram regarding a conventional configuration; and
[0071] FIG. **15** is an explanatory diagram regarding a conventional configuration.
DESCRIPTION OF THE EMBODIMENTS
[0072] Hereinafter, a description will be given, with reference to the drawings, of various exemplary embodiments (examples), features, and aspects of the present disclosure. However, the sizes, materials, shapes, their relative arrangements, or the like of constituents described in the embodiments may be appropriately changed according to the configurations, various conditions, or the like of apparatuses to which the disclosure is applied. Therefore, the sizes, materials, shapes, their relative arrangements, or the like of the constituents described in the embodiments do not intend to limit the scope of the disclosure to the following embodiments.
[0073] In the following description, the term “electrophotographic image forming apparatus” (hereinafter, also referred to as “image forming apparatus”) means an arrangement by which images are formed on a sheet-like recording medium such as paper, using an electrophotographic image forming system. Examples of an image forming apparatus include a photocopier, a facsimile apparatus, a printer (laser beam printer, light-emitting diode (LED) printer, or the like), an all-in-

one machine (multifunction printer), and so forth. Also, in the following description, the term “developing apparatus” means a unit that is used in the aforementioned image forming apparatus, and that has process means that acts on a photosensitive member (e.g., a charging member, a developing member, a cleaning member, and so forth), for example. Also, in the following description, the term “cartridge” is a unit that is detachably attachable to the aforementioned image forming apparatus, and that has means that acts on the photosensitive member (e.g., charging member, developing member, toner, cleaning member, and so forth).

First Embodiment

[0074] A first embodiment of the present invention will be described below with reference to the drawings. In the first embodiment, an image forming apparatus regarding which four toner cartridges and four process cartridges are detachably attachable is exemplified as the image forming apparatus. Note that the number of toner cartridges and process cartridges that are attached to the image forming apparatus is not limited to this number. The number thereof is to be set as appropriate in accordance with necessity. Also, a laser beam printer is exemplified as a form of the image forming apparatus in the embodiments described below.

Schematic Configuration of Image Forming Apparatus

[0075] FIG. **2** is a cross-sectional schematic view of the image forming apparatus M according to the first embodiment. FIG. **2** also illustrates a cross-section of cartridges **100** that are attached to the image forming apparatus M. The image forming apparatus M is a four-color full-color laser printer using an electrophotographic process, and performs color image formation on a recording medium S. The image forming apparatus M is an electrophotographic recording system, in which the cartridges are removably attached to an apparatus main body **170** of the image forming apparatus M to form color images on the recording medium S.

[0076] The image forming apparatus M is equipped with a front door **11** that is capable of opening and closing with respect to the apparatus main body **170**. Hereinafter, with respect to the image forming apparatus M, a side on which the front door **11** is provided will be referred to as “front face side”, and a face on a side opposite from the front face as “rear face side”. Also, a left side in a width direction (lateral direction) when viewing the image forming apparatus M from the front face will be referred to as “non-driving side”, and the right side as “driving side”. Also, an upper side when viewing the image forming apparatus M installed on a level face from the front will be referred to as “upper side”, and a lower side as “lower side”. FIG. **2** is a cross-sectional view, viewing the image forming apparatus M from the non-driving side, in which a near side from the plane of the drawing is the non-driving side of the image forming apparatus M, a far side from the plane of the drawing is the driving side of the image forming apparatus M, the right side in the plane of the drawing is the front face side of the image forming apparatus M, and the left side in the plane of the drawing is the rear face side of the image forming apparatus M.

[0077] Hereinafter, of depthwise directions of the image forming apparatus M, a direction toward the rear face is an X**1** direction, and a direction toward the front face is an X**2** direction. Also, an upward direction is a Z**1** direction, and a downward direction is a Z**2** direction. Also, of the width directions of the image forming apparatus M, a direction toward the driving side is a Y**1** direction, and a direction toward the non-driving side is a Y**2** direction. In the first embodiment, the directions X**1**, X**2**, Y**1**, and Y**2** are substantially level directions.

[0078] Four cartridges **100** (**100**Y, **100**M, **100**C, **100**K) are disposed in the apparatus main body **170**, arrayed in a substantially level direction (depthwise direction). The four cartridges **100** (**100**Y, **100**M, **100**C, **100**K) each have electrophotographic process mechanisms that are the same as each other, with colors of the developer (hereinafter referred to as “toner”) differing. Hereinafter, when there is no need to distinguish among the four cartridges **100** (**100**Y, **100**M, **100**C, **100**K), these may simply be referred to as “cartridge **100**”.

[0079] Rotational driving force is transmitted from a drive output portion of the apparatus main body **170** to the cartridges **100**. The drive output portion of the apparatus main body **170** will be described in detail later. Also, bias voltage (charging bias, developing bias, and so forth) is supplied from the apparatus main body **170** to the cartridges **100**.

[0080] The cartridge **100**Y accommodates yellow (Y) toner, and forms yellow toner images on a surface of a photosensitive drum **104**. The cartridge **100**M accommodates magenta (M) toner, and forms magenta toner images on the surface of the photosensitive drum **104**. The cartridge **100**C accommodates cyan (C) toner, and forms cyan toner images on the surface of the photosensitive drum **104**. The cartridge **100**K accommodates black (K) toner, and forms black toner images on the surface of the photosensitive drum **104**.

[0081] A laser scanner unit **14** is provided upward from the four cartridges **100** (**100**Y, **100**M, **100**C, **100**K), as exposing means. This laser scanner unit **14** outputs laser light U in accordance with image information.

[0082] A intermediate transfer unit **12** is provided downward from the four cartridges **100** (**100**Y, **100**M, **100**C, **100**K), as a transfer member. This intermediate transfer unit **12** has a driving roller **12***e*, a turn roller **12***c*, and a tension roller **12***b* over which a transfer belt **12***a*, which is flexible, is run.

[0083] The photosensitive drum **104**, serving as an image bearing member on which electrostatic latent images are formed, is provided in each cartridge **100**. The photosensitive drum **104** is a rotating body that is configured so as to be rotatable about a rotation axis line that is parallel to the width direction. A lower face of the photosensitive drum **104** is in contact with an upper face of the transfer belt **12***a*. A contact portion of the photosensitive drum **104** and the transfer belt **12***a* is a primary transfer portion. A primary transfer roller **12***d* is provided corresponding to each cartridge **100**, on an inner side of the transfer belt **12***a* at a position facing the photosensitive drum **104**.

[0084] A secondary transfer roller **6** abuts the driving roller **12***e* across the transfer belt **12***a*. A contact portion of the transfer belt **12***a* and the secondary transfer roller **6** is a secondary transfer portion.

[0085] A sheet feed unit **4** is provided downward from the intermediate transfer unit **12**. The sheet feed unit **4** has a sheet feed tray **4***a* that accommodates the recording medium S stacked therein, and a sheet feed roller **4***b*.

[0086] A fixing apparatus **7** and a sheet discharging apparatus **8** are provided to the upper left within the apparatus main body **170** in FIG. **2**. An upper face of the apparatus main body **170** makes up a sheet discharge tray **13**.

Image Forming Operations

[0087] Next, image forming operations for the image forming apparatus M to form full-color images will be described. Upon image forming operations being started, first, the photosensitive drums **104** of the four cartridges **100** (**100**Y, **100**M, **100**C, **100**K) are rotationally driven clockwise in FIG. **2**, at a predetermined speed. Also, the driving roller **12***e* is rotationally driven by a motor (omitted from illustration) of the apparatus main body **170**, and accordingly the transfer belt **12***a* moves in a direction of an arrow D in FIG. **2**.

[0088] Next, in each of the cartridges **100**, a charging roller **105** uniformly charges the surface of the photosensitive drum **104** to a predetermined polarity and potential. The laser scanner unit **14** scans and exposes the surface of each of the photosensitive drums **104** by the laser light U in accordance with image signals of the respective colors. Accordingly, electrostatic latent images in accordance with the image signals of corresponding colors are formed on the surfaces of the respective photosensitive drums **104** that are image bearing members. The electrostatic latent images that are formed are then developed by developing rollers **106** that are rotationally driven at a predetermined speed. The developing rollers **106** are toner bearing members that bear toner for developing the electrostatic latent images.

[0089] The electrophotographic image forming process operations described above form a yellow toner image corresponding to a yellow component of the full-color image on the photosensitive drum **104** of the cartridge **100**Y. Primary transfer of this toner image onto the transfer belt **12***a* is then performed.

[0090] In the same way, a magenta toner image corresponding to a magenta component of the full-color image is formed on the photosensitive drum **104** of the cartridge **100**M. Primary transfer of this toner image is then performed, so as to be superimposed onto the yellow toner image that is already transferred onto the transfer belt **12***a*.

[0091] In the same way, a cyan toner image corresponding to a cyan component of the full-color image is formed on the photosensitive drum **104** of the cartridge **100**C. Primary transfer of this toner image is then performed, so as to be superimposed onto the yellow and magenta toner images that are already transferred onto the transfer belt **12***a*.

[0092] In the same way, a black toner image corresponding to a black component of the full-color image is formed on the photosensitive drum **104** of the cartridge **100**K. Primary transfer of this toner image is then performed, so as to be superimposed onto the yellow, magenta, and cyan toner images that are already transferred onto the transfer belt **12***a*.

[0093] Thus, a four-color full-color unfixed toner image of yellow, magenta, cyan, and black is formed on the transfer belt **12***a*.

[0094] Meanwhile, the recording medium S is separated and fed at a predetermined control timing, one sheet at a time. The recording medium S is guided to the secondary transfer portion, which is the contact portion of the secondary transfer roller **6** and the transfer belt **12***a*, at a predetermined control timing. Thus, the four-color superimposed toner image on the transfer belt **12***a* is progressively transferred together onto a face of the recording medium S, in a process of the recording medium S being conveyed to the secondary transfer portion.

[0095] The recording medium S onto which the toner image is transferred then has the toner image fixed by fixing means that the fixing apparatus **7** is equipped with, and is discharged to the sheet discharge tray **13** by the sheet discharging apparatus **8**.

Cartridge Attaching/Detaching Configuration

[0096] Next, a cartridge tray (hereinafter referred to as “tray”) **171** that supports the cartridges **100** will be described in detail, with reference to FIGS. **3** to **5**. FIG. **3** is a cross-sectional view of the image forming apparatus M in a state in which the tray **171** is drawn out to the outside of the apparatus main body **170**, in a state in which the front door **11** is open. FIG. **4** is a cross-sectional view of the image forming apparatus M in a state in which the tray **171** is located outside of the apparatus main body **170** in a state in which the front door **11** is open, and toner cartridges **101** are removed from process cartridges **102**. FIG. **5** is a cross-sectional view of the image forming apparatus M in a state in which the tray **171** is located outside of the apparatus main body **170** in a state in which the front door **11** is open, and the process cartridges **102** are removed from the tray **171**.

[0097] As illustrated in FIGS. **3** to **5**, the tray **171** is movable in the X**1** direction (direction of pushing in) and the X**2** direction (direction of drawing out) with respect to the apparatus main body **170**, which are substantially level directions. That is to say, the tray **171** is provided so as to be capable of being drawn out and pushed in with respect to the apparatus main body **170**, and the tray **171** is configured to be capable of moving in the substantially level directions in a state in which the apparatus main body **170** is installed on a level face. In other words, the tray **171** is configured so as to be movable between an outside position of being located outside of the apparatus main body **170** (position illustrated in FIGS. **3** to **5**), and an inside position of being located inside of the apparatus main body **170** (position illustrated in FIG. **2**). When located at the outside position, the tray **171** is located above the front door **11** that is in an open state.

[0098] The four cartridges **100** (**100**Y, **100**M, **100**C, **100**K) are made up of the process cartridges **102** (**102**Y, **102**M, **102**C, **102**K) and the toner cartridges **101** (**101**Y, **101**M, **101**C, **101**K), respectively. As illustrated in FIG. **4**, the toner cartridges **101** are configured to be removable from the process cartridges **102**.

[0099] When the toner is gone from a toner cartridge **101**, and printing cannot be performed, a user

removes the toner cartridge **101**. The user then attaches a new toner cartridge **101**, by matching a toner-cartridge-side attaching portion **125** of the new toner cartridge **101** and a process-cartridge-side attaching portion **117** of the process cartridge **102**. The user thereafter pushes the tray **171** into the apparatus main body **170** in the X**1** direction and closes the front door **11**, whereby the image forming apparatus M is in a state capable of executing image forming operations.
[0100] Also, the tray **171** has attaching portions **171***a* (**171***a*Y, **171***a*M, **171***a*C, **171***a*K) from which the process cartridges **102** are removable at the outside position, as illustrated in FIG. **5**. When the photosensitive drums, the developing rollers, or the like, of the image forming apparatus M deteriorate and printing quality can no longer be maintained, the user removes the process cartridges **102** from the attaching portions **171***a*, and attaches new process cartridges **102** to the attaching portions **171***a*. The user then attaches the toner cartridges **101**, pushes the tray **171** into the apparatus main body **170** and closes the front door **11**, whereby the image forming apparatus M is in a state capable of image formation.
[0101] Note that the process cartridges **102** may be integrally formed with the apparatus main body **170**. In a case of a configuration in which the process cartridges **102** are non-removable from the apparatus main body **170** in this way, just the toner cartridges **101** are removable from the apparatus main body **170** (process cartridges **102**), as illustrated in FIG. **4**.
[0102] In the first embodiment, the intermediate transfer unit **12** is configured to be capable of moving up and down, by the front door **11** and a link mechanism that is omitted from illustration. Opening the front door **11** causes the intermediate transfer unit **12** to descend in the Z**2** direction, and the photosensitive drum **104** and the transfer belt **12***a* are distanced from each other. Conversely, closing the front door **11** causes the intermediate transfer unit **12** to rise in the Z**1** direction, and the photosensitive drum **104** and the transfer belt **12***a* come into contact with each other. Thus, the tray **171** can move the process cartridges **102** inside and outside of the apparatus main body **170**, without the photosensitive drum **104** and the transfer belt **12***a* being in contact.
[0103] As described above, by using the tray **171**, a plurality of the cartridges **100** can be moved inside and outside of the apparatus main body **170** together, thus enabling the toner cartridges **101** and the process cartridges **102** to be easily replaced.
Detailed Configuration of Cartridge
[0104] Next, the configuration of the cartridge **100** will be described in detail with reference to FIGS. **1**, **6**, and **7**. FIG. **1** is a cross-sectional view illustrating an inner configuration of the cartridge **100**. FIG. **6** is a perspective view illustrating an external appearance of the cartridge **100**. FIG. **7** is a schematic cross-sectional view of a developer container **110** and a toner container **120** that accommodate (hold) toner T, which are provided in the cartridge **100**. In the first embodiment, the four cartridges **100** (**100**Y, **100**M, **100**C, **100**K) have electrophotographic process mechanisms that are the same as each other, with colors of the toner T accommodated therein, and amounts of toner T filled therein, differing from each other. The configuration of the cartridge **100** in the following description is the same for all four cartridges **100** (**100**Y, **100**M, **100**C, **100**K).
[0105] Inside of the cartridge **100** is filled with toner T that is the developer. In the cartridge **100**, the toner T is accommodated in two accommodating portions, which are a first accommodating portion **110***a* and a second accommodating portion **120***a*. In FIG. **1** and other drawings, the toner T accommodated in the cartridge **100** is illustrated with black dots, and a wavy line indicating an upper face of the toner T with which the cartridge **100** is filled. The cartridge **100** is made up of the process cartridge **102** (first cartridge) and the toner cartridge **101** (second cartridge). The process cartridge **102** is configured to be detachably attachable to the image forming apparatus M, develops latent images on the photosensitive drum **104** in a stable manner with the toner T that is supplied thereto, which is transferred onto the intermediate transfer unit **12**. The toner cartridge **101** is configured to be detachably attachable to the process cartridge **102**, and supplies the toner T to the process cartridge **102**.
[0106] The process cartridge **102** has the photosensitive drum **104** that is a photosensitive member

that is rotatable, a container **130** that holds the photosensitive drum **104**, and the charging roller **105** that is rotatable. The charging roller **105** is biased against the photosensitive drum **104** by a charging spring **105***a*, and is configured to be rotatable about the rotation axis line that is parallel to the width direction of the image forming apparatus M, in the same way as the photosensitive drum **104**. A charge is input from the apparatus main body **170** to the charging roller **105** by a charging contact point that is omitted from illustration, whereby the surface of the photosensitive drum **104** is uniformly charged.

[0107] A drum coupling **160** for transmitting driving force to the photosensitive drum **104** is provided on one end of each photosensitive drum **104** in a longitudinal direction. Note that the photosensitive drum **104** is attached to the image forming apparatus M such that the longitudinal direction thereof is parallel to the width direction of the image forming apparatus M. The drum coupling **160** engages a drum drive coupling **180** (illustrated in FIG. **3**) serving as a drum drive output portion of the apparatus main body **170**. Driving force of a driving motor (omitted from illustration) of the apparatus main body **170** is transmitted to the photosensitive drum **104**, whereby the photosensitive drum **104** is rotated in a direction of arrow A in FIG. **1** (clockwise direction).

[0108] Also, the process cartridge **102** is a developing apparatus that has the developer container **110** in which the first accommodating portion **110***a*, for accommodating the toner T that is developer supplied from the toner cartridge **101**, is formed. A replenishing opening **110***c* for supplying the toner T from the developer container **110** toward a developing chamber **118** is formed in a lower face **110***b* that is located on a lower side out of inner faces of the developer container **110**, in an orientation in which the cartridge **100** is attached to the apparatus main body **170**. A supply roller **107** and the developing roller **106** serving as a developing member are provided inside of the developing chamber **118**. The developing chamber **118** is located below the developer container **110** (first accommodating portion **110***a*), in an orientation in which the cartridge **100** is attached to the apparatus main body **170**.

[0109] A lid **114** is attached to the developer container **110**. The lid **114** is located on an upper side of the developer container **110** in an orientation in which the cartridge **100** is attached to the apparatus main body **170**, and forms the first accommodating portion **110***a* along with the developer container **110**. The lid **114** is located between the first accommodating portion **110***a* and the second accommodating portion **120***a* (details to be described later) of the toner container **120**, and makes up a boundary portion that partitions between the first accommodating portion **110***a* and the second accommodating portion **120***a*. That is to say, a face **114***a* of the lid **114** that is a face that faces the first accommodating portion **110***a* is an upper face of the first accommodating portion **110***a*, and makes up an inner wall face of the first accommodating portion **110***a*. A channel (omitted from illustration) for supplying the toner T from the second accommodating portion **120***a* to the first accommodating portion **110***a* is formed in the lid **114**.

[0110] A magnetism sensing sensor **115** that is capable of sensing magnetism is provided inside of the lid **114**. The magnetism sensing sensor **115** is disposed at position that is different from the channel provided to the lid **114** through which the toner T passes, in a direction parallel to a first rotation axis line **111***a*.

[0111] A first rotation shaft **111** for stirring the toner T within the developer container is provided in the first accommodating portion **110***a*. The first rotation shaft **111** is configured to be rotatable about the first rotation axis line **111***a* that is parallel to the width direction (rotation axis line of the photosensitive drum **104**) of the image forming apparatus M. A first stirring sheet **109** that supplies the toner T within the developer container **110** to the supply roller **107**, and a first flexible sheet **112**, are attached to the first rotation shaft **111**. A first magnet **113** that is a first magnetized member is attached to a distal end portion of the first flexible sheet **112**.

[0112] The first stirring sheet **109** is a sheet-like flexible member (third flexible member). The first stirring sheet **109** extends in a direction intersecting the first rotation axis line **111***a*, and has a fixed end **109***a* (third fixed end) that is fixed to the first rotation shaft **111**, and a free end **109***b* (third free end). The first stirring sheet **109** extends over substantially the entire region of the first accommodating portion **110***a* in the direction parallel to the first rotation axis line **111***a*.

[0113] In a state in which the first stirring sheet **109** is not flexing, a distance from the first rotation axis line **111***a* (center of rotation) of the first rotation shaft **111** to the free end **109***b* of the first stirring sheet **109** is a length R**1**. FIG. **7** illustrates an imaginary circle VC**1** of which the length R**1** is the radius, by a dash-double-dot line. The imaginary circle VC**1** corresponds to the path over which a distal end of the first stirring sheet **109** passes when the first rotation shaft **111** is rotated in a state in which the first stirring sheet **109** does not flex. The length R**1** is set to be greater than a distance from the first rotation axis line **111***a* to the lower face **110***b* of the developer container **110** that makes up a replenishing opening **110***c*. According to such a configuration, the first stirring sheet **109** conveys the toner T while coming into contact with the lower face **110***b* of the developer container **110** in conjunction with rotation of the first rotation shaft **111**. The toner T is thus fed out from the developer container **110** toward the supply roller **107** via the replenishing opening **110***c*.

[0114] As illustrated in FIG. **6**, a developer coupling **161** is provided on one end of the cartridge **100** in the longitudinal direction. When the process cartridge **102** is attached to the apparatus main body **170**, the developer coupling **161** and a developer driving coupling **185** that is provided to the apparatus main body **170** (illustrated in FIG. **3**) are engaged. Engagement of the developer coupling **161** and the developer driving coupling **185** transmits driving force of a driving motor (omitted from illustration) of the apparatus main body **170** to the first rotation shaft **111**. The first rotation shaft **111** to which driving force is transmitted is rotated in a direction of arrow B in FIG. **1** and other drawings (counterclockwise direction).

[0115] The toner T supplied to the supply roller **107** is supplied to the developing roller **106**. A developing blade **108** then regulates the toner T on the developing roller **106** to a uniform thickness. Thus, toner can be stably supplied to the photosensitive drum **104** with which the developing roller **106** is in contact.

[0116] The first flexible sheet **112** is a sheet-like flexible member (first flexible member). The first flexible sheet **112** extends in a direction intersecting the first rotation axis line **111***a*, and has a first fixed end **112***a* that is fixed to the first rotation shaft **111**, and a first free end **112***b*. In a direction parallel to the first rotation axis line **111***a*, the width of the first flexible sheet **112** is smaller than the width of the first stirring sheet **109**. The longitudinal direction of the first flexible sheet **112** is a direction extending from the first fixed end **112***a* to the first free end **112***b*.

[0117] In a state in which the first flexible sheet **112** is not flexing, a distance from the first rotation axis line **111***a* (center of rotation) of the first rotation shaft **111** to the first free end **112***b* of the first flexible sheet **112** is a length R**2**. FIG. **7** illustrates an imaginary circle VC**2** of which the length R**2** is the radius, by a dash-double-dot line. The imaginary circle VC**2** corresponds to the path over which the distal end of the first flexible sheet **112** passes when the first rotation shaft **111** is rotated in a state in which the first flexible sheet **112** does not flex. The length R**2** is set to be smaller than a distance from the first rotation axis line **111***a* to the face **114***a* of the lid **114** (upper face of first accommodating portion **110***a*). Accordingly, when the first rotation shaft **111** is rotating, the first flexible sheet **112** does not come into contact with the lid **114** that is the boundary portion of the first accommodating portion **110***a* and the second accommodating portion **120***a*.

[0118] The first rotation shaft **111** has a first face **111***b* to which the fixed end **109***a* (third fixed end) of the first stirring sheet **109** is fixed, and a second face **111***c* to which the first fixed end **112***a* of the first flexible sheet **112** is fixed. In the first embodiment, the first face **111***b* and the second face **111***c* face opposite directions from each other. Also, a direction in which the first stirring sheet **109** protrudes from the first rotation shaft **111** and a direction in which the first flexible sheet **112** protrudes from the first rotation shaft **111** are opposite directions from each other.

[0119] The first magnet **113** is provided in a region between the first fixed end **112***a* and first free end **112***b* of the first flexible sheet **112**. More specifically, the first magnet **113** is provided at a position that is closer to the first free end **112***b* than the first fixed end **112***a* of the first flexible sheet **112**. The first magnet **113** is provided on a face of the first flexible sheet **112** that is fixed to the first rotation shaft **111** and that faces a downstream side in a direction of rotation when the first rotation shaft **111** rotates in the direction of the arrow B. The first magnet **113** is disposed so as not to come into contact with the lid **114**, the same as with the first flexible sheet **112** when the first rotation shaft **111** rotates.

[0120] The toner cartridge **101** has the toner container **120** in which is formed the second accommodating portion **120***a* for accommodating toner T to be supplied to the process cartridge **102**. A supply opening (omitted from illustration) for supplying toner T from the toner container **120** toward the developer container **110** is formed in a lower face **120***b* that makes up an inner wall face of the second accommodating portion **120***a*, in an orientation in which the cartridge **100** is attached to the apparatus main body **170**. Also, in a state in which the toner cartridge **101** is attached to the process cartridge **102**, the second accommodating portion **120***a* is aligned with the first accommodating portion **110***a* in a first direction D**1**. In the first embodiment, the first direction D**1** is a direction that is orthogonal to the Y**1** (Y**2**) direction, and that intersects the Z**1** (Z**2**) direction and the X**1** (X**2**) direction. An angle formed between the first direction D**1** and the Z**1** direction is smaller than an angle formed between the first direction D**1** and the X**1** direction.

[0121] A second rotation shaft **121** for stirring the toner T within the toner container **120** is provided in the toner container **120**. The second rotation shaft **121** is configured to be rotatable about a second rotation axis line **121***a* that is parallel to the width direction of the image forming apparatus M. A second stirring sheet **122** that supplies the toner T within the toner container **120** to the developer container **110**, and a second flexible sheet **123** are attached to the second rotation shaft **121**. A second magnet **124** that is a second magnetized member is attached to a distal end portion of the second flexible sheet **123**.

[0122] The second stirring sheet **122** is a sheet-like flexible member (fourth flexible member). The second stirring sheet **122** extends in a direction intersecting the second rotation axis line **121***a*, and has a fixed end **122***a* (fourth fixed end) that is fixed to the second rotation shaft **121**, and a free end **122***b* (fourth free end). The second stirring sheet **122** extends over substantially the entire region of the second accommodating portion **120***a* in a direction parallel to the second rotation axis line **121***a*.

[0123] In a state in which the second stirring sheet **122** is not flexing, a distance from the second rotation axis line **121***a* (center of rotation) of the second rotation shaft **121** to the free end **122***b* of the second stirring sheet **122** is a length R**3**. FIG. **7** illustrates an imaginary circle VC**3** of which the length R**3** is the radius, by a dash-double-dot line. The imaginary circle VC**3** corresponds to the path over which a distal end of the second stirring sheet **122** passes when the second rotation shaft **121** is rotated in a state in which the second stirring sheet **122** does not flex. The length R**3** is set to be greater than a distance from the second rotation axis line **121***a* to the lower face **120***b* of the toner container **120** in which the supply opening is formed, and to the supply opening formed in the lower face **120***b*. Accordingly, the second stirring sheet **122** conveys the toner T while coming into contact with the lower face **120***b* of the toner container **120** in conjunction with rotation of the second rotation shaft **121**. The toner T is thus fed out from the toner container **120** toward the developer container **110** via the supply opening.

[0124] The cartridge **100** has a driving transmission mechanism that transmits drive force from the process cartridge **102** to the second rotation shaft **121** of the toner cartridge **101**. Upon driving force being transmitted from the process cartridge **102** to the second rotation shaft **121**, the second rotation shaft **121** rotates in a direction of arrow C illustrated in FIG. **1** and other drawings. The second stirring sheet **122** then conveys the toner T, and the toner T is supplied to the process cartridge **102** via the supply opening.

[0125] The second flexible sheet **123** is a sheet-like flexible member (second flexible member). The second flexible sheet **123** extends in a direction intersecting the second rotation axis line **121***a*, and has a second fixed end **123***a* that is fixed to the second rotation shaft **121**, and a second free end **123***b*. In a direction parallel to the second rotation axis line **121***a*, the width of the second flexible sheet **123** is smaller than the width of the second stirring sheet **122**. The longitudinal direction of the second flexible sheet **123** is a direction extending from the second fixed end **123***a* to the second free end **123***b*.

[0126] In a state in which the second flexible sheet **123** is not flexing, a distance from the second rotation axis line **121***a* (center of rotation) of the second rotation shaft **121** to the second free end **123***b* of the second flexible sheet **123** is a length R**4**. FIG. **7** illustrates an imaginary circle VC**4** of which the length R**4** is the radius, by a dash-double-dot line. The imaginary circle VC**4** corresponds to the path over which the distal end of the second flexible sheet **123** passes when the second rotation shaft **121** is rotated in a state in which the second flexible sheet **123** does not flex. The length R**4** is set to be smaller than a distance from the second rotation axis line **121***a* to the lower face **120***b* of the toner container **120** in which the supply opening is formed, and to the supply opening formed in the lower face **120***b*. Accordingly, when the second rotation shaft **121** is rotating, the second flexible sheet **123** does not come into contact with the lower face **120***b*.

[0127] The second rotation shaft **121** has a third face **121***b* to which the fixed end **122***a* (fourth fixed end) of the second stirring sheet **122** is fixed, and a fourth face **121***c* to which the second fixed end **123***a* of the second flexible sheet **123** is fixed. In the first embodiment, the third face **121***b* and the fourth face **121***c* face opposite directions from each other. Also, a direction in which the second stirring sheet **122** protrudes from the second rotation shaft **121** and a direction in which the second flexible sheet **123** protrudes from the second rotation shaft **121** are opposite directions from each other.

[0128] The second magnet **124** is provided in a region between the second fixed end **123***a* and the second free end **123***b* of the second flexible sheet **123**. More specifically, the second magnet **124** is provided at a position that is closer to the second free end **123***b* than the second fixed end **123***a* of the second flexible sheet **123**. The second magnet **124** is provided on a face of the second flexible sheet **123** that is fixed to the second rotation shaft **121** and that is a face that faces the downstream side in the direction of rotation when the second rotation shaft **121** rotates in the direction of the arrow C. The second magnet **124** is disposed so as not to come into contact with the lower face **120***b* when the second rotation shaft **121** rotates, the same as with the second flexible sheet **123**.

[0129] Note that while the cartridge is configured as an integral all-in-one cartridge in the first embodiment, application of the present invention is not limited to such a configuration. For example, a configuration may be made in which the toner cartridge that holds the toner and the process cartridge that has the developing roller, the supply roller, the drum, the charging roller, and the cleaning member, may each be independently detachably attachable to the apparatus main body. Alternatively, a form may be made in which the cartridge is made up of a developing unit that has the developing roller, the supply roller, and the toner, and a drum unit that holds the photosensitive drum, the charging roller, and the cleaning member. Also, a configuration may be made in which a configuration that is equivalent to the process cartridge **102** is not removable from the apparatus main body, and only a cartridge that is equivalent to the toner cartridge **101** is detachably attachable. Further, an image forming apparatus may be configured such that configurations that are equivalent to the process cartridge **102** and the toner cartridge **101** are not removable from the apparatus main body.

Remaining Toner Amount Sensing Method

[0130] Next, a sensing method of remaining toner amount by the image forming apparatus M will be described with reference to FIGS. **8**A to **8**C, FIGS. **9**A to **9**C, FIGS. **10**A to **10**C, and FIGS. **11** and **12**. Upon printing operations being executed in the image forming apparatus M, toner is consumed, and the amount of toner within the cartridge **100** decreases. If printing operations happen to be executed in a state in which there is scant toner within the cartridge **100**, sufficient toner may not be printed on printed articles, resulting in defective printing. Accordingly, it is desirable for the image forming apparatus M to be configured such that a situation in which there is a scant amount of toner within the cartridge **100** can be comprehended, and the user can be provided with a report thereof before the toner runs out. Thus, the image forming apparatus M is configured to be capable of sensing the amount of toner in the cartridge **100**.

[0131] As described above, the first flexible sheet **112** is attached to the first rotation shaft **111** of the process cartridge **102**, and the first magnet **113** serving as the first magnetized member is attached to the distal end of the first flexible sheet **112**. Also, the second flexible sheet **123** is attached to the second rotation shaft **121** of the toner cartridge **101**, and the second magnet **124** serving as the second magnetized member is attached to the distal end of the second flexible sheet **123**. Note that while a magnet is used as the magnetized member in the first embodiment, this can be substituted with another member that is magnetized (emits magnetism).

[0132] Also, as described above, the magnetism sensing sensor **115** that senses magnetism within a sensing-capable range is attached to the lid **114** of the process cartridge **102**. The magnetism sensing sensor **115** is provided between the first rotation axis line **111***a* and the second rotation axis line **121***a* in the first direction D**1**. That is to say, the magnetism sensing sensor **115** is located in a region Ar**1** (see FIG. **7**) between an imaginary line Ln**2** that passes through the first rotation axis line **111***a* and an imaginary line Ln**3** that passes through the second rotation axis line **121***a*, which are perpendicular to an imaginary line Ln**1** that passes through the first rotation axis line **111***a* and the second rotation axis line **121***a*. Note that the imaginary line Ln**2** and the imaginary line Ln**3** are imaginary lines that extend in the second direction D**2** that is perpendicular to the first direction D**1** and the Y**1** (Y**2**) direction. In the first embodiment, the magnetism sensing sensor **115** is disposed between the first accommodating portion **110***a* formed in the developer container **110** of the process cartridge **102** and the second accommodating portion **120***a* formed in the toner container **120** of the toner cartridge **101**.

[0133] As described above, the first magnet **113** and the second magnet **124** are each provided on a flexible sheet of which an end portion is fixed to the rotation shaft. Accordingly, a distance between the first magnet **113** and the magnetism sensing sensor **115** changes in conjunction with rotation of the first rotation shaft **111**, and a distance between the second magnet **124** and the magnetism sensing sensor **115** changes in conjunction with rotation of the second rotation shaft **121**. According to such a configuration, the first magnet **113** and the second magnet **124** are disposed so as to be capable of entering and exiting a sensing-capable range of the magnetism sensing sensor **115**. When the first magnet **113** and the second magnet **124** enter the sensing-capable range of the magnetism sensing sensor **115**, in conjunction with rotation of the first rotation shaft **111** and the second rotation shaft **121**, the magnetism sensing sensor **115** senses magnetism. In the first embodiment, the remaining toner amount is acquired on the basis of results of sensing by the magnetism sensing sensor **115**. The magnetism sensing sensor **115** can be disposed on the imaginary line Ln**1** that passes through the first rotation axis line **111***a* and the second rotation axis line **121***a*, for example.

[0134] The magnetism sensing sensor **115** performs detection while distinguishing between magnetism of the first magnet **113** and of the second magnet **124**, and accordingly the phases of the first rotation shaft **111** and the second rotation shaft **121** are controlled such that the first magnet **113** and the second magnet **124** do not enter the sensing-capable range of the magnetism sensing sensor **115** at the same time. A method of controlling the phases of the first rotation shaft **111** and the second rotation shaft **121** by a control portion in the first embodiment will be described below.

[0135] FIGS. **8**A to **8**C are explanatory diagrams illustrating a relation of phases of the first rotation shaft **111** and the second rotation shaft **121**, and are schematic cross-sectional view of the cartridge **100**. FIG. **8**A illustrates the way in which a new toner cartridge **101** is attached to the process cartridge **102**. When replacing the toner cartridge **101**, the phase of the first rotation shaft **111** provided in the process cartridge **102** is controlled by the magnetism sensing sensor **115** so as to be constantly unchanged. In the first embodiment, the phase of the first rotation shaft **111** is controlled such that the first face **111***b* to which the first stirring sheet **109** is fixed faces upward, and the first flexible sheet **112** faces downward, in an orientation in which the cartridge **100** is attached to the apparatus main body **170**.

[0136] Also, in the toner cartridge **101** that is to be newly attached, the phase of the second rotation shaft **121** is set so as to avert a situation in which the first magnet **113** and the second magnet **124** are located in the sensing-capable range of the magnetism sensing sensor **115** at the same time. Specifically, the phase of the second rotation shaft **121** is controlled such that the third face **121***b* to which the second stirring sheet **122** is fixed faces upwards and the fourth face **121***c* to which the second flexible sheet **123** is fixed faces downwards, in an orientation in which the cartridge **100** is attached to the apparatus main body **170**. That is to say, when the toner cartridge **101** is attached to the process cartridge **102**, the phase of the first rotation shaft **111** and the phase of the second rotation shaft **121** are the same.

[0137] Further, the same gear is used as a gear for rotationally driving the first rotation shaft **111** and a gear for rotationally driving the second rotation shaft **121**, and the revolutions of the first rotation shaft **111** and the second rotation shaft **121** are constantly the same. That is to say, the phases of the first rotation shaft **111** and the second rotation shaft **121** are controlled such that the second face **111***c* (first plane) of the first rotation shaft **111** and the fourth face **121***c* (second plane) of the second rotation shaft **121** are parallel to each other, and also face the same direction. According to such a configuration, when the first rotation shaft **111** rotates and the first magnet **113** nears the magnetism sensing sensor **115**, for example, the second magnet **124** is at a position distanced from the magnetism sensing sensor **115** as compared to the first magnet **113**. FIG. **8**B illustrates the way in which the first magnet **113** nears the magnetism sensing sensor **115**, and the second magnet **124** is being distanced from the magnetism sensing sensor **115**. Further continuing rotation of the first rotation shaft **111** and the second rotation shaft **121** from the state in FIG. **8**B distances the first magnet **113** from the magnetism sensing sensor **115**, and the second magnet **124** nears the magnetism sensing sensor **115**. FIG. **8**C illustrates the way in which the first magnet **113** is distanced from the magnetism sensing sensor **115**, and the second magnet **124** nears the magnetism sensing sensor **115**. In this way, according to the first embodiment, the phases of the first rotation shaft **111** and the second rotation shaft **121** are controlled such that the timings at which the first magnet **113** and the second magnet **124** near the magnetism sensing sensor **115** are made to be different from each other. That is to say, when the first magnet **113** is located in the sensing-capable range of the magnetism sensing sensor **115**, the second magnet **124** is located outside of this sensing-capable range, and when the second magnet **124** is located in the sensing-capable range of the magnetism sensing sensor **115**, the first magnet **113** is located outside of this sensing-capable range. Thus, according to such a configuration, a situation in which the first magnet **113** and the second magnet **124** are both located in the sensing-capable range of the magnetism sensing sensor **115** at the same time can be averted, and magnetism of both the first magnet **113** and of the second magnet **124** can be detected by the single magnetism sensing sensor **115**.

[0138] Whether magnetism sensed by the magnetism sensing sensor **115** is from the first magnet **113** or from the second magnet **124** may be distinguished by managing a relation between the phases of the first rotation shaft **111** and the second rotation shaft **121**, and point in time, for example. The magnet that is at a position near to the magnetism sensing sensor **115** can be found from the point in time at which the magnetism sensing sensor **115** senses magnetism, and the phases of the first rotation shaft **111** and the second rotation shaft **121**. Alternatively, the initial phases of the first rotation shaft **111** and the second rotation shaft **121** when a new toner cartridge **101** is attached may be set as described above, and distinguishing may be performed in which the magnetism that is sensed first is that of the first magnet **113**, and the magnetism that is sensed second is that of the second magnet **124**.

[0139] Next, an acquisition method of the amount of toner will be described. The acquisition method of the amount of toner in the cartridge **100** is the same for all four cartridges **100** (**100**Y,

100M, **100**C, **100**K).

[0140] First, the acquisition method of amount of toner in the toner cartridge **101** will be described. FIGS. **9**A to **9**C are explanatory diagrams regarding the acquisition method of the amount of toner in the second accommodating portion **120***a* of the toner cartridge **101**, and illustrate the cartridge **100** in an orientation of being attached to the apparatus main body **170**. FIG. **11** is a graph illustrating a relation between the amount of toner in the second accommodating portion **120***a* of the toner cartridge **101**, and amount of time of sensing by the magnetism sensing sensor **115**.

[0141] FIG. **9**A illustrates a situation in which the toner T accommodated in the second accommodating portion **120***a* of the toner container **120** is plentiful (the amount of toner is great), and the highest face of the toner T is upward from the second rotation shaft **121**. The second flexible sheet **123** can be made of a polyethylene terephthalate (PET) or polyphenylene sulfide (PPS) sheet that is 50 μm thick, for example. In a state in which the toner T in the second accommodating portion **120***a* is plentiful, when the second rotation shaft **121** rotates in the direction of the arrow C, the second flexible sheet **123** rotates while flexing under resistance from the toner T. The shortest distance between the second magnet **124** and the magnetism sensing sensor **115** when the second magnet **124** is nearest to the magnetism sensing sensor **115** in such a state is distance L**1**.

[0142] FIG. **9**B illustrates a situation in which the amount of toner accommodated in the second accommodating portion **120***a* of the toner container **120** is less than that in the case of FIG. **9**A. When the amount of toner in the second accommodating portion **120***a* decreases and the resistance of the toner T that the second flexible sheet **123** is subjected to is reduced, the amount of flexing of the second flexible sheet **123** decreases. Then, when the amount of flexing of the second flexible sheet **123** when the second flexible sheet **123** passes beneath the second rotation shaft **121** decreases, the shortest distance between the second magnet **124** and the magnetism sensing sensor **115** becomes smaller.

[0143] FIG. **9**C illustrates a situation in which the amount of toner accommodated in the second accommodating portion **120***a* of the toner container **120** is less than that in the case of FIG. **9**B. In such a case, the resistance of the toner T that the second flexible sheet **123** is subjected to is reduced even further, the amount of flexing of the second flexible sheet **123** decreases even further. Accordingly, the shortest distance between the second magnet **124** and the magnetism sensing sensor **115** when the second magnet **124** is nearest to the magnetism sensing sensor **115** becomes a distance L**2** that is smaller than the distance L**1** (L**1**>L**2**).

[0144] As described above, the shortest distance between the second magnet **124** and the magnetism sensing sensor **115** gradually becomes smaller from a certain point in conjunction with the reduction in the amount of toner in the second accommodating portion **120***a*. The shortest distance becoming smaller means that the amount of time that the second magnet **124** is in the sensing-capable range of the magnetism sensing sensor **115** becomes longer in each rotation of the second rotation shaft **121**, for example. FIG. **11** shows the relation between the amount of time of sensing by the magnetism sensing sensor **115** and the amount of toner in the second accommodating portion **120***a*.

[0145] As shown in FIG. **11**, when the amount of toner is no less than a predetermined amount (portion D in FIG. **11**), the amount of time of sensing by the magnetism sensing sensor **115** is constant. At this time, in a case in which the amount of flexing of the second flexible sheet **123** is great, and the second magnet **124** does not enter into the sensing-capable range of the magnetism sensing sensor **115**, the amount of time of sensing by the magnetism sensing sensor **115** will be constant at zero. The reason is that in a case in which a great amount of toner T is accommodated in the second accommodating portion **120***a* as illustrated in FIG. **9**A, even if the amount of toner decreases somewhat, the amount of flexing of the second flexible sheet **123** when the second magnet **124** nears the magnetism sensing sensor **115** hardly changes at all.

[0146] Conversely, when the amount of toner is no more than the predetermined amount (portion E in FIG. **11**), the amount of time of sensing by the magnetism sensing sensor **115** becomes longer in conjunction with the amount of toner decreasing. The reason is that in a case in which the amount of toner in the second accommodating portion **120***a* is small, as illustrated in FIG. **9**C, the amount of flexing of the second flexible sheet **123** is reduced, and the second magnet **124** comes nearer to the magnetism sensing sensor **115**. At this time, the amount of toner in the second accommodating portion **120***a* and the amount of time of sensing by the magnetism sensing sensor **115** are proportionate. Accordingly, the image forming apparatus M can acquire (estimate) the amount of toner in the second accommodating portion **120***a* on the basis of the amount of time of sensing by the magnetism sensing sensor **115**, by a toner amount acquisition portion. For the toner amount acquisition portion, a control portion made up of a central processing unit (CPU) or the like, for example, may be used. The toner amount acquisition portion can be provided at any position, such as the apparatus main body **170** of the image forming apparatus M, the cartridge **100**, or the like, for example.

[0147] According to the cartridge **100** configured as described above, the image forming apparatus M can acquire the amount of toner in the second accommodating portion **120***a* on the basis of the results of sensing by the magnetism sensing sensor **115**. The user can then be notified, by notification means such as a monitor or the like that the image forming apparatus M is equipped with, that the amount of toner remaining in the toner cartridge **101** is either scant or is zero, and thereby prompt replacement of the toner cartridge **101**. Note that as illustrated in FIGS. **9**A to **9**C, when the second magnet **124** is located in the sensing-capable range of the magnetism sensing sensor **115**, the first magnet **113** is distanced from the magnetism sensing sensor **115**, and is located at a position away from this sensing-capable range.

[0148] Next, the acquisition method of the amount of toner in the process cartridge **102** will be described. FIGS. **10**A to **10**C are explanatory diagrams regarding the acquisition method of the amount of toner in the first accommodating portion **110***a* of the process cartridge **102**, and illustrate the cartridge **100** in an orientation of being attached to the apparatus main body **170**. FIG. **12** is a graph showing a relation between the amount of toner in the first accommodating portion **110***a* of the process cartridge **102**, and amount of time of sensing by the magnetism sensing sensor **115**.

[0149] FIG. **10**A illustrates a situation in which the toner T accommodated in the first accommodating portion **110***a* of the developer container **110** is plentiful (the amount of toner is great), and the highest face of the toner T is located above the first rotation shaft **111**. The first flexible sheet **112** can be made of a PET or PPS sheet that is 50 μm thick, for example. In a state in which the toner T in the first accommodating portion **110***a* is plentiful, when the first rotation shaft **111** rotates in the direction of the arrow B, the first flexible sheet **112** rotates while flexing under resistance from the toner T. The shortest distance between the first magnet **113** and the magnetism sensing sensor **115** when the first magnet **113** is nearest to the magnetism sensing sensor **115** in such a state is distance L**3**.

[0150] FIG. **10**B illustrates a situation in which the amount of toner accommodated in the first accommodating portion **110***a* of the developer container **110** is less than that in the case of FIG. **10**A. When the amount of toner in the first accommodating portion **110***a* decreases and the resistance of the toner T that the first flexible sheet **112** is subjected to is reduced, the amount of flexing of the first flexible sheet **112** decreases. When the amount of flexing of the first flexible sheet **112** when the first flexible sheet **112** passes beneath the first rotation shaft **111** decreases, the shortest distance between the first magnet **113** and the magnetism sensing sensor **115** becomes smaller.

[0151] FIG. **10**C illustrates a situation in which the amount of toner accommodated in the first accommodating portion **110***a* of the developer container **110** is less than that in the case of FIG. **10**B. In such a case, the resistance of the toner T that the first flexible sheet **112** is subjected to is reduced even further, the amount of flexing of the first flexible sheet **112** decreases even further. Accordingly, the shortest distance between the first magnet **113** and the magnetism sensing sensor

115 when the first magnet **113** is nearest to the magnetism sensing sensor **115** becomes a distance LA that is smaller than the distance L**3** (L**3**>L**4**).

[0152] As described above, the shortest distance between the first magnet **113** and the magnetism sensing sensor **115** gradually becomes smaller, in conjunction with the decrease in the amount of toner in the first accommodating portion **110***a*. The shortest distance becoming smaller means that the amount of time that the first magnet **113** is located in the sensing-capable range of the magnetism sensing sensor **115** becomes longer in each rotation of the first rotation shaft **111**, for example. FIG. **12** shows a relation between the amount of time of sensing by the magnetism sensing sensor **115** and the amount of toner in the first accommodating portion **110***a*.

[0153] As shown in FIG. **12**, when the amount of toner is no less than a predetermined amount (portion F in FIG. **12**), the amount of time of sensing by the magnetism sensing sensor **115** becomes longer in conjunction with the reduction in the amount of toner. The reason is that when the amount of toner in the first accommodating portion **110***a* decreases, the flexing amount of the first flexible sheet **112** decreases, and the first magnet **113** comes nearer to the magnetism sensing sensor **115**, as illustrated in FIGS. **10**A to **10**C. At this time, the amount of toner in the first accommodating portion **110***a* and the amount of time of sensing by the magnetism sensing sensor **115** are proportionate. Accordingly, the image forming apparatus M can acquire (estimate) the amount of toner in the first accommodating portion **110***a* on the basis of the amount of time of sensing of the magnetism sensing sensor **115** by the toner amount acquisition portion.

[0154] Conversely, when the amount of toner is no more than the predetermined amount (portion G in FIG. **12**), the amount of time of sensing by the magnetism sensing sensor **115** is constant. The reason is that in a case in which the amount of toner T accommodated in the first accommodating portion **110***a* is small, as illustrated in FIG. **10**C, even if the amount of toner decreases somewhat, the amount of flexing of the first flexible sheet **112** when the first magnet **113** nears the magnetism sensing sensor **115** hardly changes at all.

[0155] According to the cartridge **100** configured as described above, the image forming apparatus M can acquire the amount of toner in the first accommodating portion **110***a* on the basis of the results of sensing by the magnetism sensing sensor **115**. The user can then be notified, by notification means such as a monitor or the like that the image forming apparatus M is equipped with, that the amount of toner remaining in the toner cartridge **101** is either scant or is zero, and thereby prompt replacement of the toner cartridge **101** or the cartridge **100**. Note that as illustrated in FIGS. **10**A to **10**C, when the first magnet **113** is located in the sensing-capable range of the magnetism sensing sensor **115**, the second magnet **124** is distanced from the magnetism sensing sensor **115**, and is located at a position away from this sensing-capable range.

[0156] Note that when the amount of toner is no more than the predetermined amount (portion G in FIG. **12**), the amount of time of sensing the first magnet **113** by the magnetism sensing sensor **115** does not change even if the amount of toner decreases. The image forming apparatus M may be configured such that the user is notified of the amount of toner in the process cartridge **102** at the point that the amount of toner reaches this predetermined amount. Alternatively, a configuration may be made in which, when the amount of toner is no more than the predetermined amount (portion G in FIG. **12**), the amount of toner is acquired (estimated) by another method, such as, for example, estimating the amount of toner consumption from an image pattern. Generally, toner amount sensing methods in which the toner consumption amount is estimated from image patterns tend to accumulate error. However, by combining with the toner amount sensing method that uses magnetism, all that is necessary is to perform calculations regarding just the small amount of toner that is running out in the developer container **110**, on the basis of the toner consumption amount, and accordingly the amount of toner can be sensed with high precision until the amount of toner becomes zero.

Method for Improving Position Precision of Magnetism Sensing Sensor

[0157] In the first embodiment, a configuration is described in which the magnetism sensing sensor

115 is fixed to the lid **114**, but application of the present invention is not limited to such a configuration. The attachment position and attachment method of the magnetism sensing sensor **115** can be changed as appropriate, in accordance with the sensitivity of the magnetism sensing sensor **115**, the magnetic forces of the first magnet **113** and the second magnet **124**, the positional relation of the first accommodating portion **110***a* and the second accommodating portion **120***a*, and so forth. An attachment method that is capable of improving positional precision of the magnetism sensing sensor **115** will be described below, as an example of the attachment method of the magnetism sensing sensor **115**.

[0158] FIGS. **13**A and **13**B are cross sectional views of the cartridge **100** illustrating an example of an attachment method of the magnetism sensing sensor **115**. FIG. **13**A illustrates the toner cartridge **101** in a state of having been removed from the process cartridge **102**. FIG. **13**B illustrates the toner cartridge **101** in a state of having been attached to the process cartridge **102**. In this example, a sensor spring **116** is disposed between the magnetism sensing sensor **115** and the toner cartridge **101**. The sensor spring **116** is a biasing member that biases the magnetism sensing sensor **115** toward the process cartridge **102** (first accommodating portion **110***a*) in a state in which the toner cartridge **101** is attached (assembled) to the process cartridge **102**. According to such a configuration, the magnetism sensing sensor **115** is positioned in close contact with the toner cartridge **101**, and accordingly precision relating to the relative position thereof as to the magnets is improved. Thus, small-sized and inexpensive magnets can be selected.

[0159] As described above, according to the above-described configuration, in a small-sized color laser beam printer (LBP) configuration having a paired replenishing configuration divided into a developer container and a toner container, the amount of toner in both the developer container side and the toner container side can be sensed by a single magnetism sensing sensor. Furthermore, a cartridge and an image forming apparatus that perform remaining amount sensing can be provided with a simple configuration and also with conserved space.

[0160] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0161] This application claims the benefit of Japanese Patent Application No. 2024-027430, filed on Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A cartridge detachably attachable to an image forming apparatus, the cartridge comprising: a first accommodating portion configured to accommodate toner; a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line; a first flexible member extending in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end; a second accommodating portion configured to accommodate toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction; a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line; a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end of the second flexible member, including the second free end; and a magnetism sensing sensor provided between the first rotation axis line and the second rotation axis line in the first direction, and configured to sense magnetism in a sensing-capable range.

2. The cartridge according to claim 1, wherein the second accommodating portion is located above the first accommodating portion, in an orientation in which the cartridge is attached to the image forming apparatus.

3. The cartridge according to claim 1, wherein the second magnetized member is located outside of the sensing-capable range in a case in which the first magnetized member is located inside of the sensing-capable range, and wherein the first magnetized member is located outside of the sensing-capable range in a case in which the second magnetized member is located inside of the sensing-capable range.

4. The cartridge according to claim 1, further comprising: a control portion controlling phases of the first rotation shaft and the second rotation shaft, wherein the first rotation shaft has a first plane to which the first flexible member is fixed, and the second rotation shaft has a second plane to which the second flexible member is fixed, and wherein the control portion controls the phases of the first rotation shaft and the second rotation shaft such that the first plane of the first rotation shaft and the second plane of the second rotation shaft are parallel to each other and also face a same direction.

5. The cartridge according to claim 1, further comprising: a boundary portion that separates the first accommodating portion and the second accommodating portion from each other, wherein the magnetism sensing sensor is provided at the boundary portion.

6. The cartridge according to claim 5, wherein the boundary portion constitutes an upper face of the first accommodating portion and a lower face of the second accommodating portion, in an orientation in which the cartridge is attached to the image forming apparatus.

7. The cartridge according to claim 5, wherein, in a state in which the first flexible member is not flexing, a distance from the first rotation axis line to the first free end is smaller than a distance from the first rotation axis line to the boundary portion, and wherein, in a state in which the second flexible member is not flexing, a distance from the second rotation axis line to the second free end is smaller than a distance from the second rotation axis line to the boundary portion.

8. The cartridge according to claim 7, wherein the first magnetized member is provided at a position that is closer to the first free end of the first flexible member than to the first fixed end thereof, and wherein the second magnetized member is provided at a position that is closer to the second free end of the second flexible member than to the second fixed end thereof.

9. The cartridge according to claim 1, wherein the magnetism sensing sensor is disposed on an imaginary line that passes through the first rotation axis line and the second rotation axis line, as viewed from a direction parallel to the first rotation axis line.

10. The cartridge according to claim 1, wherein the cartridge includes a first cartridge and a second cartridge detachably attached to the first cartridge, the first cartridge having the first accommodating portion, the first rotation shaft, the first flexible member, and the first magnetized member, the second cartridge having the second accommodating portion, the second rotation shaft, the second flexible member, and the second magnetized member.

11. The cartridge according to claim 10, further comprising: a biasing member that biases the magnetism sensing sensor toward the first accommodating portion in a state in which the second cartridge is attached to the first cartridge.

12. The cartridge according to claim 5, further comprising: a developing chamber inside which a toner bearing member configured to bear toner is provided, wherein an opening, through which toner supplied from the first accommodating portion to the developing chamber passes, is formed in the first accommodating portion.

13. The cartridge according to claim 12, further comprising: a third flexible member extending in a direction intersecting the first rotation axis line, the third flexible member having a third fixed end that is fixed to the first rotation shaft, and a third free end opposite to the third fixed end in the direction intersecting the first rotation axis line; and a fourth flexible member extending in a direction intersecting the second rotation axis line, the fourth flexible member having a fourth fixed end that is fixed to the second rotation shaft, and a fourth free end opposite to the fourth fixed end in the direction intersecting the second rotation axis line.

14. The cartridge according to claim 13, wherein, in a state in which the third flexible member is not flexing, a distance from the first rotation axis line to the third free end is greater than a distance from the first rotation axis line to the opening, and wherein, in a state in which the fourth flexible member is not flexing, a distance from the second rotation axis line to the fourth free end is greater than a distance from the second rotation axis line to an opening which is formed in the second accommodating portion, and through which toner supplied to the first accommodating portion passes.

15. An image forming apparatus, comprising: an apparatus main body; and a cartridge configured to be detachably attachable to the apparatus main body, the cartridge including a first accommodating portion configured to accommodate toner, a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line; a first flexible member extending in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end; a second accommodating portion configured to accommodate toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction; a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line; a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end of the second flexible member, including the second free end; and a magnetism sensing sensor provided between the first rotation axis line and the second rotation axis line in the first direction, and configured to sense magnetism in a sensing-capable range.

16. The image forming apparatus according to claim 15, further comprising: a toner amount acquisition portion configured to acquire an amount of toner in the first accommodating portion and an amount of toner in the second accommodating portion, on the basis of amount of time over which the magnetism sensing sensor has sensed magnetism.

17. An image forming apparatus, comprising: an image bearing member on which an electrostatic latent image is formed; a toner bearing member configured to bear toner for developing the electrostatic latent image; a first accommodating portion that accommodates the toner to be supplied to the toner bearing member; a first rotation shaft that is provided inside of the first accommodating portion and that is configured to be rotatable about a first rotation axis line; a first flexible member that extends in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; a first magnetized member that is provided in a region of the first flexible member between the first fixed end of the first flexible member and the first free end thereof, including the first free end; a second accommodating portion that accommodates toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction; a second rotation shaft that is provided inside of the second accommodating portion and that is configured to be rotatable about a second rotation axis line; a second flexible member that extends in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; a second magnetized member that is provided in a region of the second flexible member between the second fixed end of the second flexible member and the second free end thereof, including the second free end; and a magnetism sensing sensor that is provided between the first rotation axis line and the second rotation axis line in the first direction, and that senses magnetism in a sensing-capable range.

18. A developing apparatus, comprising: a developing member; a first accommodating portion configured to accommodate toner; a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line; a first flexible member extending in a direction intersecting the first rotation axis line, the first flexible member having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line; a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end; and a magnetism sensing sensor configured to sense magnetism in a sensing-capable range, wherein the developing apparatus is configured such that a cartridge is detachably attachable, the cartridge including a second accommodating portion accommodating toner to be supplied to the first accommodating portion, a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line, a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line, and a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end of the second flexible member, including the second free end, wherein the first accommodating portion is aligned with the second accommodating portion in a first direction, in an orientation in which the cartridge is attached to the developing apparatus, and wherein the cartridge is attached such that the magnetism sensing sensor is located between the first rotation axis line and the second rotation axis line in the first direction.

19. A toner cartridge configured to be detachably attachable to a developing apparatus that includes a developing member, a first accommodating portion configured to accommodate toner, a first rotation shaft provided inside of the first accommodating portion and configured to be rotatable about a first rotation axis line, a first flexible member extending in a direction intersecting the first rotation axis line and having a first fixed end that is fixed to the first rotation shaft, and a first free end opposite to the first fixed end in the direction intersecting the first rotation axis line, a first magnetized member provided in a region of the first flexible member between the first fixed end and the first free end of the first flexible member, including the first free end and a magnetism sensing sensor configured to sense magnetism in a sensing-capable range, the toner cartridge comprising: a second accommodating portion configured to accommodate toner to be supplied to the first accommodating portion, the second accommodating portion being aligned with the first accommodating portion in a first direction in an orientation in which the toner cartridge is attached to the developing apparatus; a second rotation shaft provided inside of the second accommodating portion and configured to be rotatable about a second rotation axis line; a second flexible member extending in a direction intersecting the second rotation axis line, the second flexible member having a second fixed end that is fixed to the second rotation shaft, and a second free end opposite to the second fixed end in the direction intersecting the second rotation axis line; and a second magnetized member provided in a region of the second flexible member between the second fixed end and the second free end thereof of the second flexible member, including the second free end, wherein the magnetism sensing sensor is attached to the developing apparatus so as to be located between the first rotation axis line and the second rotation axis line in the first direction.